BLINK Multimodal Large Language Models Can See but Not Perceive (2404.12390v4)

Abstract

BLINK Multimodal Large Language Models Can See but Not Perceive (2404.12390v4), p.1

> because they resist mediation through natural language.

BLINK tasks are difficult for LMMs. 14 classic computer vision tasks are segregated into 3807 multiple choice questions and recent proprietary models also break under these questions.

Conclusion

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> We conduct extensive analysis, measuring the effect of converting images to dense captions, visual prompting, self-consistency, analyzing the capabilities of specialist models, and conducting error analysis.

Process followed by authors

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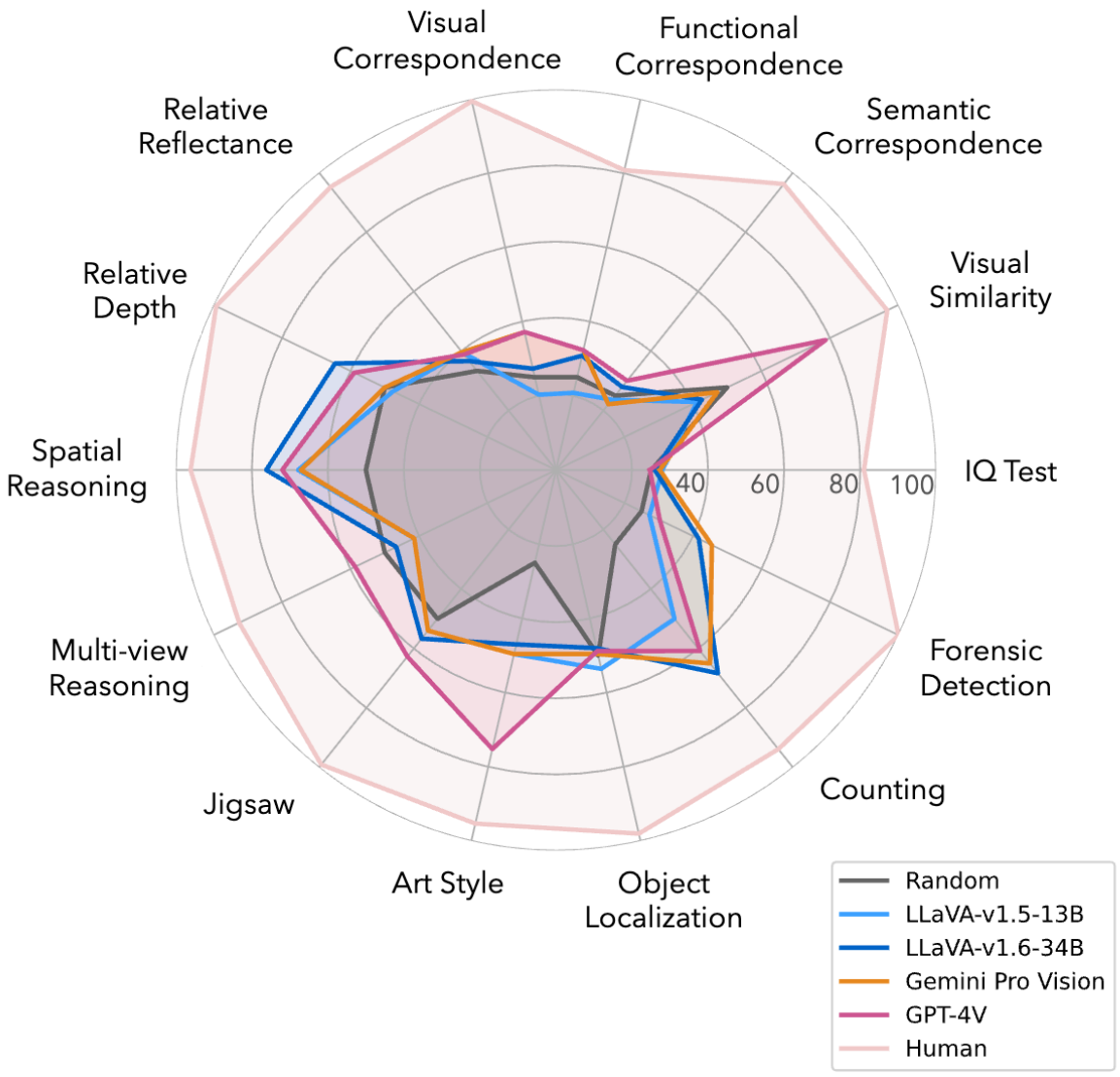
> We highlight that specialist computer vision models are performing much better than GPT-4V and Gemini on Blink, shedding light on the possibility that multimodal LLMs may have big progress on these tasks

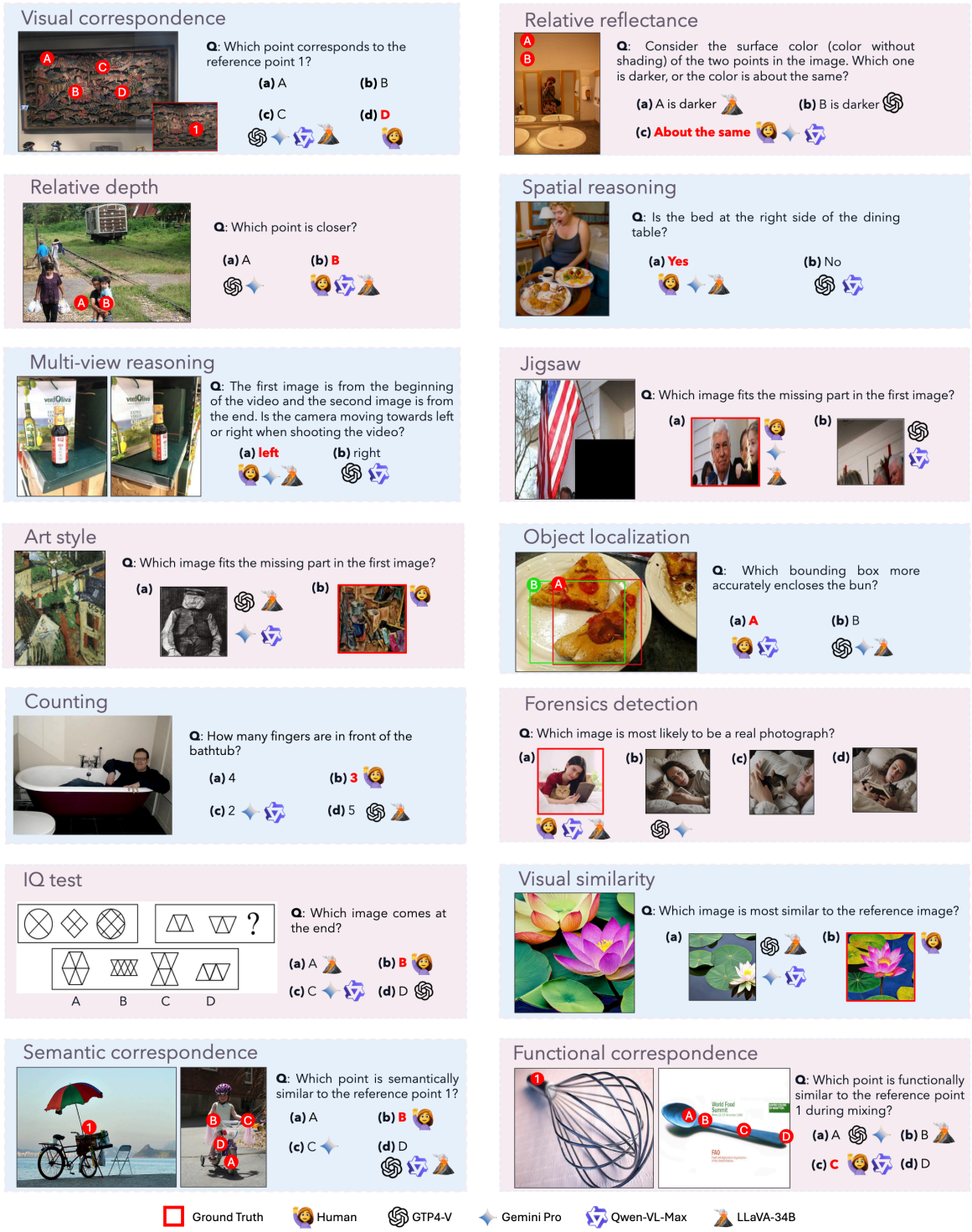
Specialist models are found to perform better

Figures and tables

BLINK provides number of guidance points, like bboxes, reference points. BLINK focuses more on visual perception tasks.



Models can be seen performing well below humans on visual tasks. Generally performance is good on spatial reasoning and counting.



BLINK has 14 categories of questions. However, the answers are closed form and limited to 4 options.

Task	Vis.Corr.	Depth	Multi-view	Sem.Corr.	Forensic	Reflect.
Random	25.00	50.00	50.00	25.00	25.00	33.33
Human	99.56	99.59	92.10	94.60	100.00	99.63
Gemini Pro	42.44	40.32	44.36	26.62	50.76	45.52
GPT-4V	33.72	59.68	55.64	28.78	34.09	38.81
Specialist	DIFT [70]	DepthAnything [83]	LoFTR [68]	DIFT [70]	DIRE [79]	Ordinal Shading [14]
	96.51	97.58	90.22	71.22	68.94	77.61

Even propreitary models perform worse than humans.

Introduction

BLINK Multimodal Large Language Models Can See but Not Perceive (2404.12390v4), p.2

> This might be because many traditional computer vision tasks resist mediation through natural language, due to the inherent imprecision of language (e.g., it is challenging to precisely pinpoint a spatial keypoint through language).

Important statement

BLINK Multimodal Large Language Models Can See but Not Perceive (2404.12390v4), p.2

> Our experiments show that this conflation reductively evaluates perception as a dense captioning task. In other words, by replacing the image with a task-agnostic dense caption, our experiments show that a “blind” GPT-4 performs well on these “multimodal tasks”.

Benchmarks do not really test vision capabilities. The text description of the image would suffice to answer the benchmark questions of previous benchmarks

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> he image tasks are meticulously selected such that they are difficult to solve by reducing the evaluation using dense captioning; instead, the models must perceive the contents of the image(s) to answe

BLINK Multimodal Large Language Models Can See but Not Perceive (2404.12390v4), p.4

> We observe the paradox that while these problems are easy for humans (95.70% average accuracy), they are extremely hard for existing machinery – even GPT-4V model can only achieve 51.26% accuracy on average, which is 44.44% worse than humans, and 13.17% better than random guessing.

Finding

BLINK Multimodal Large Language Models Can See but Not Perceive (2404.12390v4), p.4

> We also experiment with specialist vision models and find that they perform much better than multimodal LLMs. F

What are these specialists models?

BLINK Multimodal Large Language Models Can See but Not Perceive (2404.12390v4), p.4

> Our findings indicate that the perceptual abilities of multimodal LLMs have been previously overestimated. Furthermore, these models may benefit from integrating insights from specialized models that excel in these areas.

Since other benchmarks focus more in VQA and blink relies more on visual tasks, authors claim previous benchmarks overestimated vision capabilities.

Results and Experiments

BLINK Multimodal Large Language Models Can See but Not Perceive (2404.12390v4), p.5

> To ensure that one can effectively measure what Multimodal LLMs can or cannot perceive, we carefully select 14 tasks (see §3.2 for the full list) that are difficult to solve by reducing the evaluation into text-only questions using dense captioning.

BLINK tasks are difficult to solve by text-only evaluation. For other Benchmarks, this can be done.

BLINK Multimodal Large Language Models Can See but Not Perceive (2404.12390v4), p.5

> hey range from low-level pattern matching (e.g., visual correspondence) to mid-level spatial reasoning (e.g., relative depth), and up to high-level visual understanding (e.g., visual similarity). This variety allows for a systematic exploration of Multimodal LLMs' capabilities across different perceptual complexity layers.

Multiple tasks to develop fine-grained analysis of the capabilities of the LMMs.

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> granularity, ranging from pixels (e.g., relative reflectance) to patches (e.g., jigsaw) and extending to the full image (e.g., forensic detection), enabling us to evaluate models' proficiency in observing at various scales.

Scale is tested on image-level. Model scale is potentially limited.

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> Visual commonsense" that does not require domain knowledge

Questions of BLINK do not require experts. Humans can also solve them with everyday knowledge.

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> Blink features a heterogeneous questionanswering format, wherein both questions and choices can be presented as text or images.

Also contains interleaved format.

Overall

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> Overall performance: As shown in Table 1, the mean accuracy of 7B and 13B open-source Multimodal LLMs hover around 35–42%, which is similar to random guess (38.09%). The most proficient open-source model, LLaVA-v1.6-34B, achieves an accuracy of 45.05%. Even the most advanced models, GPT-4V and Gemini Pro and Claude 3 OPUS, achieve accuracies of only 51.26%, 45.72%, and 44.11% respectively. Their performance are merely 13.17%, 7.63% and 6.02% better than random guessing and lag behind human performance by 44.44%, 49.98% and 51.59%. Notably, for certain tasks such as jigsaw, semantic correspondence, multiview reasoning, object localization, and relative reflectance, some multimodal LLMs even underperform compared to random guessing. Some qualitative results are shown in Figure 5.

Open source and proprietary models perform only slightly better than random baseline.

BLINK Multimodal Large Language Models Can See but Not Perceive (2404.12390v4), p.11

> ultimodal LLMs perform relatively better on spatial reasoning, art style, and counting tasks, in which they are much better than random guessing.

Scores range from 46 to 77% on Spatial reasoning task.

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> models also demonstrate some capability in relative depth and forensics detection. Overall, they are doing relatively well on mid-level perception tasks.

Continuation of better performance

Analysis

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> If using captions along with text-only LLMs yields performance comparable to or surpassing that achieved through the integration of images with multimodal LLMs, then the perception demands of that benchmark are primarily confined to recognition only.

1. The authors make a good observation here. If performance on text-only questions is better than or equal to the corresponding image-text questions, then the benchmark is confined to recognition only.
2. Another perspecitve: This can also be indicative of modality gap. If text performance is better than image-text performance, then a modality gap exists.

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> he performance decrease observed in Blink suggests the necessity for advanced perceptual abilities beyond what is currently attainable with general captions.

MMBench and MMMU are recognition benchmarks, as their text caption only scores are better than or equal to the image-text scores. They are also better than baseline. However, text-only scores on BLINK fail to meet the baseline.

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> Specialists can serve as a proxy upper bound of how good multimodal LLMs could be.

Specialists provide the ceiling for performance measures.

BLINK Multimodal Large Language Models Can See but Not Perceive (2404.12390v4), p.15

> Error analysis of GPT-4V: We randomly sampled 140 error instances made by GPT-4V on Blink, 10 per task, and meticulously examined them. The most common types of errors are: Hallucinate fine-grained patterns and attributes (24.2%): the model hallucinates the nuanced details of objects. This error is most common for relative reflectance, forensics detection, and jigsaw tasks. Hallucinate visual prompt locations (20.0%): the circle location described by the model is

> wrong. This is common for visual correspondence and relative depth tasks. Other errors include Failures on capturing overall setting or style (8.6%), and Failures on grounding an object (5.7%). More details are in Appendix C.3.

Authors found hallucination as the primary error.